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Elimination kinetics of perfluorohexanoic acid in humans and comparison with mouse, rat and monkey

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HIGHLIGHTS

- Elimination half-life of PFHxA determined from blood of ski wax technicians.
- Geomean human elimination half-life of PFHxA is 32 d (range 14–49 d).
- Elimination half-life of PFHxA in mammalian species is proportional to body weight.
- Rate of PFHxA elimination from mammals is useful for subsequent risk assessment.

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ABSTRACT

Major fluorinated chemical manufacturers have developed new short-chain per- and polyfluorinated substances with more favorable environmental, health and safety profiles. This study provides the first evaluation of the elimination half-life of perfluorohexanoic acid (PFHxA) from the blood of humans. PFHxA biomonitoring data were obtained from a recently published study of professional ski wax technicians. These data were analyzed to provide estimates of the apparent half-life of PFHxA from humans, and comparisons were made with kinetic studies of PFHxA elimination from mice, rats and monkeys. The apparent elimination half-life of PFHxA in highly exposed humans ranged between 14 and 49 d with a geomean of 32 d. The half-lives of PFHxA in mice, rats, monkeys and humans were proportional to body weight with no differences observed between genders, indicating similar volumes of distribution and similar elimination mechanisms among mammalian species. Compared to long-chain perfluoroalkyl acid analogs, PFHxA is rapidly cleared from biota. The consistent weight-normalized elimination half-lives for PFHxA in mammalian species indicates that results obtained from animal models are suitable for establishment of PFHxA benchmark dose and reference dose hazard endpoints for use in human risk assessments.

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1. Introduction

Long-chain perfluoroalkyl acids (PFAAs) such as perfluorooctane sulfonate (PFOS), perfluorohexane sulfonate (PFHxS), and perfluorooctanoic acid (PFOA) have been widely found in the environment and in humans from both their direct use and from precursor substances that degrade to form them (OECD, 2010; Buck et al., 2011). Leading global fluorinated chemical manufacturers have committed to work toward the elimination of long-chain PFAAs (i.e., perfluoroalkyl sulfonic acids with ⩾6 perfluorinated carbons or perfluoroalkyl carboxylic acids with ⩾7 perfluorinated carbons) and their potential precursors (USEPA, 2010). New products have been developed based on shorter-chain per- and poly-fluoroalkyl substances that have a more favorable environmental, health and safety profile (Chengelis et al., 2009a; Loveless et al., 2009; Olsen et al., 2009; Iwai, 2011) and have been approved by regulators (Ritter, 2010; Cronkhite, 2012). A key attribute of these new short-chain substances is their rapid rate of elimination from biota including humans which results in very low or non-detectable concentrations in blood and renders these substances non-bioaccumulative.

PFAAs such as perfluorocarboxylic acids (PFCAs, e.g., C_nF_{2n+1}-COOH) and perfluoroalkane sulfonates (PFSAs, e.g., $C_nF_{2n+1}SO_3^-$) are generally recognized to be stable under normal environmental and biological conditions (Liou et al., 2010). The widespread detection of persistent long-chain PFCAs and PFSAs such as PFOA, PFOS and PFHxS in human blood reflects the net result of the competitive rate of uptake via various routes of exposure and the

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subsequent rate of bioelimination. PFOA, PFOS and PFHxS are reported to have geomean elimination half-lives of 3.5 years, 4.8 years and 7.3 years, respectively in humans (Olsen et al., 2007). In contrast to the long systemic elimination half-lives determined for PFOA, PFOS and PFHxS, short-chain PFAAs, such as perfluorobutanoic acid (PFBA), perfluorobutane sulfonate (PFBS) and perfluorohexanoic acid (PFHxA) rapidly eliminate from blood, plasma and serum of mammals (e.g., monkey, rat and mouse) with no appreciable gender differences (Chang et al., 2008; Olsen et al., 2009; Gannon et al., 2011). Significant kinetic differences between long and short chain PFAAs in uptake and elimination have also been observed in aquatic studies with the conclusion that short-chain PFAAs neither bioconcentrate nor bioaccumulate in fish (Martin et al., 2003b,a; Conder et al., 2008).

In biomonitoring studies of the general human population, short-chain PFAAs such as PFBA, PFBS and PFHxA have been included as analytes but are rarely detected (Kato et al., 2011; Olsen et al., 2012). In more highly exposed populations (e.g., due to local contamination or workplace exposure), rapid elimination half-lives of PFBA and PFBS have been calculated (Olsen et al., 2009). Recently, a biomonitoring study was conducted on a group of ski wax technicians who were highly exposed to PFAAs and potential PFAA precursors during multiple World Cup racing seasons (Nilsson et al., 2010a,b, 2013). The analytes in this study included PFHxA, making this the first biomonitoring study useful for evaluation of the elimination half-life of this PFCA from human blood.

The objective of this paper is threefold: to provide an overview of the current human biomonitoring data for perfluorohexanoic acid (PFHxA, CAS 307-24-4) in blood, plasma and serum; to calculate the elimination half-life of PFHxA from humans; and to compare the human elimination half-life of PFHxA with the values observed in three other mammalian species: mice, rats, and monkey. This is the first study to calculate the elimination half-life of PFHxA from humans and to compare the results with other mammalian species.

2. Current observations of PFHxA in human blood

2.1. Biomonitoring of PFHxA

Over the past decade, human blood biomonitoring for perfluoroalkyl acids (PFAAs) has broadened from a predominant focus on PFOS and PFOA to evaluation of a wide range of corresponding homologs and potential precursors. Many recent studies have included analysis of short-chain PFCAs including PFHxA in human blood. Published biomonitoring results for PFHxA in human blood are summarized in Table 1. These results include concentrations observed in the general population in Europe, Asia and North America as well as concentrations reported in more highly exposed individuals who were occupationally exposed or lived in communities adjacent to manufacturing sites.

With few exceptions, PFHxA concentrations in large-scale biomonitoring surveys of the general population are low with median plasma or serum values at or below typical method limit of quantification (LOQ) values of 0.05–0.10 ng mL^{-1}. PFHxA concentrations in this range are 40–400 times lower than the biomonitoring concentrations for PFOA and PFOS which are currently reported to have median values of 4 ng mL^{-1} and 21 ng mL^{-1}, respectively, in the general population of North America (Kato et al., 2011). Biomonitoring of PFHxA has not been routinely included in the CDC National Health and Nutrition Examination Survey (NHANES) due to the low potential for detecting significant PFHxA concentrations in human blood (Calafat, A., personal communication).

In a biomonitoring study of over 60 000 members of a community adjacent to a manufacturing facility, the median PFHxA serum concentration was 0.5 ng mL^{-1} with 47% of the population having no detectable plasma concentration (i.e., less than the LOQ of 0.05 ng mL^{-1}) (Frisbee et al., 2009). In contrast, the median serum PFOA concentration in this same study was 28 ng mL^{-1} with detectable concentrations in 99.9% of the population (Frisbee et al., 2009).

2.2. Normalization of biomonitoring results

One important note should be made when comparing data from various PFCA biomonitoring studies. PFCAs primarily bind to proteins present in the plasma or serum fraction of blood with typical bound fractions of approximately 99% for PFHxA, PFOA and PFOS (Bischel et al., 2011). Since hematocrit values for adults are typically 42–45%, PFCA concentrations in plasma or serum are approximately twice as high as concentrations in blood. Therefore, comparisons of biomonitoring results for PFHxA should include appropriate consideration of the specific blood fraction included in the analysis. Fortunately, kinetic calculations within a single study are not affected by the specific blood fraction in which PFHxA is reported (i.e., in whole blood, in serum or in plasma) enabling direct comparison across studies.

3. Design and conduct of biomonitoring study of ski wax technicians

Due to the typically low PFHxA background concentrations in human blood (i.e., <LOQ or <LOD), it is difficult if not impossible to accurately calculate bioelimination half-life values from biomonitoring studies of the general population. However, a unique study of ski wax technicians who were occupationally exposed to PFAAs and potential precursors was recently published (Nilsson et al., 2010a,b, 2013). The temporal PFHxA biomonitoring data from ski wax technicians provide a valuable multi-year data set that may be used to calculate values for the apparent elimination half-life of PFHxA from humans.

3.1. Study design

Blood samples ($n = 94$) were collected from 11 male professional ski wax technicians at 11 World Cup events in Cross Country skiing during the period 2007–2011. The technicians applied fluorinated ski wax for approximately 30 h a week during each skiing season which lasts from December through March. Blood samples were taken periodically at the end of the working day during ski season and then monthly from April to August 2008 when the technicians were not working with ski wax. Before study initiation, a written informed consent was received from all participants and the study protocol was approved by the ethical vetting board of Uppsala, Sweden (Reference No. Dnr 2010/056).

3.2. Chemical analysis

Details of the analytical procedures used to detect a suite of per- and poly-fluoroalkyl chemicals including PFHxA in periodic blood samples from the ski wax technicans have previously been published (Nilsson et al., 2010b). A summary of this information, including chemical sources, extraction and analytical methodology and quality assurance/control steps are provided in Supplementary Material (SM).

Table 1
Compilation of PFHxA detections in human blood, plasma and serum, 1997–2010.

Location	Sampling date	Sample	Gender	Number samples	Percent detections	Concentration (ng mL^{-1})				Reference
						Minimum	Median	Mean	Maximum	
General environmental exposure										
Sweden	1997–2000	Blood	M F	66	8	<0.10	<0.10	–	1.6	(Kärrman, et al., 2006)
USA	2000–2001	Serum	M F	645	3	<0.10[a]	<0.10[a]	<0.10[a]	6.0	(Olsen et al., 2012)
USA	2001	Plasma	M	4	100	0.088	0.195	0.270	0.600	(Miyake et al., 2007)
Sri Lanka	2003	Serum	M	30	87	<0.020	0.07	0.30	3.30	(Guruge et al., 2005)
Poland	2003	Blood	M F	60	80	<0.002	–	.030	0.240	(Falandysz et al., 2006)
Japan	2003	Blood	M	3	33	<0.020	<0.020	<0.022	0.027	(Miyake et al., 2007)
Europe	2003	Blood	M F	47	26	0.10	0.11	–	0.24	(WWF, 2004)
China	2004	Blood	M F	30	57	<0.01	–	<0.052	0.112	(Yeung et al., 2008)
USA	2006	Plasma	M F	600	3	<0.10*	<0.10[a]	<0.10[a]	1.5	(Olsen et al., 2012)
China	2009	Serum	M F	227	27	<0.02	0.02	–	2.36	(Li et al., 2011)
USA	2009	Serum	M F	40	60	<0.001	–	0.050	0.718	(Lee and Mabury, 2011)
USA	2009	Serum	M F	>100	100	0.033	–	0.039	0.056	(Lee and Mabury, 2011)
USA	2010	Plasma	M F	600	18	<0.05[a]	<0.05[a]	<0.05[a]	0.13	(Olsen et al., 2012)
Community exposure										
USA	2005–2006	Serum	M F	66900	53	>0.25	0.5	0.9	–	(Frisbee et al., 2009)
China	2008	Blood	M F	141	99	<0.034	0.40	0.51	2.23	(Guo et al., 2011)
Occupational exposure										
Japan	2004	Blood	M	2	0	<1	<1	<1	<1	(Miyake et al., 2007)
Japan	2004	Plasma	M	2	50	0.840	<1	<1	<1	(Miyake et al., 2007)

[a] LOQ varied among groups of samples, predominant value is shown.

4. PFHxA results from biomonitoring of ski wax technicians

4.1. 6:2. FTOH monitoring in air

The technicians were exposed to various per- and poly-fluoroalkyl substances via inhalation as they used heat to apply wax to racing skis. Monitoring of the air inhaled by the technicians during the waxing process indicated the presence of relatively high concentrations of volatilized fluorotelomer alcohols (FTOHs) including 6:2 FTOH, a known metabolic precursor to PFHxA (Gannon et al., 2010; Himmelstein et al., 2012a). Ambient air concentrations of 6:2 FTOH in the breathing zone of the technicians ranged between <1.3 to 2400 ng m^{-3} with a mean of 240 ng m^{-3} (Nilsson et al., 2010b).

4.2. Metabolic fate of inhaled 6:2 FTOH

In soil and sludge, 6:2 FTOH is enzymatically converted to PFHxA with typical reaction yields of 8% (Liu et al., 2010) while yields of <1% are observed in mammalian systems (Gannon et al., 2010; Himmelstein et al., 2012b). The balance of inhaled 6:2 FTOH degrades to reactive metabolites which form conjugation products and are subsequently eliminated from the body. Based on these results, the inhalation and metabolism of 6:2 FTOH can serve as a potential exposure source of PFHxA in blood.

4.3. Monitoring of PFHxA in air

In contrast to the relatively low 6:2 FTOH concentrations in inhaled air, the concentrations of PFHxA in the breathing zone of ski wax technicians ranged between 57 and 14000 ng m^{-3} with a mean of 5300 ng m^{-3} (Nilsson et al., 2010b). Since the yield of PFHxA from 6:2 FTOH *in vivo* is <1%, the direct inhalation of PFHxA by the technicians is estimated to be 600–6000 times higher than the potential contribution of this substance arising from exposure to and metabolism of 6:2 FTOH.

4.4. PFHxA results in blood biomonitoring

Blood monitoring of the ski wax technicians occurred over a period of four years at various frequencies for each individual (Table SM1). The PFHxA blood concentrations peak during the December to March World Cup ski seasons and then rapidly decline in the spring and summer.

5. Analysis of PFHxA elimination kinetics

5.1. General elimination behavior of PFHxA

The observed temporal concentrations of PFHxA in the blood of ski wax technicians showed a consistent pattern of clearly detectable peak concentrations established during the racing season (e.g. 0.65–15.01 ng mL^{-1}) followed by elimination to values near or below the limit of quantitation (e.g. <0.05–0.14 ng mL^{-1}) in the months immediately following ski season. The cyclic pattern of exposure and uptake during ski season followed by rapid elimination during spring, summer and fall was observed repeatedly with no apparent trend of bioaccumulation after four years of seasonal exposure.

5.2. Apparent half-life versus intrinsic half-life

Ski wax technicians may experience an ongoing low level of non-occupational exposure to PFHxA from background environmental sources including metabolic conversion of potential precursors such as 6:2 FTOH in air or additional direct exposure to PFHxA from food which is the principal background exposure medium for PFCAs in the general population (Vestergren et al., 2008; Fromme et al., 2009). The PFHxA concentration decline data after the end of each ski season can be used to calculate a human elimination half-life for this substance. Due to potential ongoing non-occupational exposure to PFHxA and formation of this terminal acid from precursors during the depuration period, this calculated half-life should be referred to as an apparent half-life (Bartell, 2012). However, since ski wax technician plasma concentrations are at least two orders of magnitude greater than background concentrations, apparent and intrinsic half-life values calculated for PFHxA are expected to be similar in magnitude.

5.3. Apparent PFHxA half-life calculations for ski wax technicians

If three or more sampling times were available during the depuration periods of each ski wax technician, a first-order elimination

rate was estimated from the PFHxA blood monitoring data by regressing the experimental data to a first-order kinetic equation (see SM for details). The apparent elimination half-life is then determined from the rate constant with the usual logarithmic transformation, $HL_a = \ln(2)/k$ where HL_a is the apparent elimination half-life (d). Depuration periods for each technician were identified by locating a peak concentration during or shortly after the ski season and defining this as C_0. Subsequent blood concentrations in the spring and summer months exhibited lower PFHxA concentrations as this substance was eliminated from the blood of each technician. The available biomonitoring data for seven technicians (i.e. technicians 1–4 and 6–8) had sufficient depuration data to support reliable estimation of the apparent elimination rate (Fig. 1). The resulting geomean elimination half-life of PFHxA was 32 d with a range of 14–49 d (Table 2), indicating a relatively rapid rate of elimination of this substance in humans.

6. Comparison of PFHxA elimination kinetics in humans and comparison with mouse, rat and monkey

A number of laboratory toxicokinetic studies of PFHxA and its various salts have been conducted with CD-1® mice, Sprague-Dawley® rats and Cynomolgus monkeys using either IV or oral administration routes (Table SM2). The pK_a value for PFHxA is predicted to be approximately 0 (Rayne and Forest, 2010) and salts of perfluorohexanoate (PFHx-) with the counterions Na+, K+, Li+ and NH_4+ are expected to have similarly low pK_a values. At typical pH values in human blood, e.g. pH 7.3–7.4, PFHxA and its salts are expected to exist exclusively in anionic form resulting in high solubility as well as negligible volatility.

Based on multiple experimental studies (Chengelis et al., 2009; Gannon et al. 2011), PFHxA is rapidly absorbed, distributed and eliminated in mammalian systems. The elimination half-life of ammonium perfluorohexanoate (APFHx) dosed orally in mice was approximately 1 h (Iwai, 2011). The elimination half-life of PFHxA in rats ranged between 0.4 and 9.8 h, depending upon route of administration and number of doses administered (Himmelstein et al., 2008; Chengelis et al., 2009b; Gannon et al., 2009).

The elimination half-life of PFHxA in monkeys following a single IV dose was 2–5 h in short-term observations (i.e. 0–48 h after a single IV dose) (Chengelis et al., 2009a). A longer term study of the elimination kinetics of potassium perfluorohexanoate (KPFHx) in monkeys dosed orally resulted in a more clearly defined terminal elimination phase with elimination half-lives of 14–47 h) (Noker, 2001). In all mammalian studies, the primary route of elimination was via urine.

7. Discussion

7.1. Comparison of interspecies clearance for PFOA versus PFHxA

For a one-compartment toxicokinetic model with first-order elimination, clearance (CL, mL h^{-1} kg^{-1}) is determined by CL = k_e - $*$ V_d where k_e is the elimination half-life (h^{-1}) and V_d (mL kg^{-1}) is the volume of distribution. Since V_d values for the distribution of anionic PFCAs in rats and humans are similar (Han et al., 2012), the resulting ratios of rat/human clearance values are equivalent to inverse ratios of elimination half-life values.

Based on the results of kinetic studies of PFOA and PFHxA in humans and rats (Olsen et al., 2007; Gannon et al., 2011) and this study, PFHxA is cleared from the plasma of humans and male rats 40–80 times faster than PFOA. In contrast, the clearance of PFHxA from the plasma of female rats is less than a factor of 10 greater than the value observed for PFOA. The slow rate of PFOA elimination from humans and male rats is due to a renal resorption mechanism which effectively reduces the net secretion rate of this carboxylic acid (Han et al., 2012). This mechanism does not appear to be operative for shorter chains such as PFHxA, resulting in much faster clearance in both humans and rats. Details of these clearance calculations are provided in Table SM3.

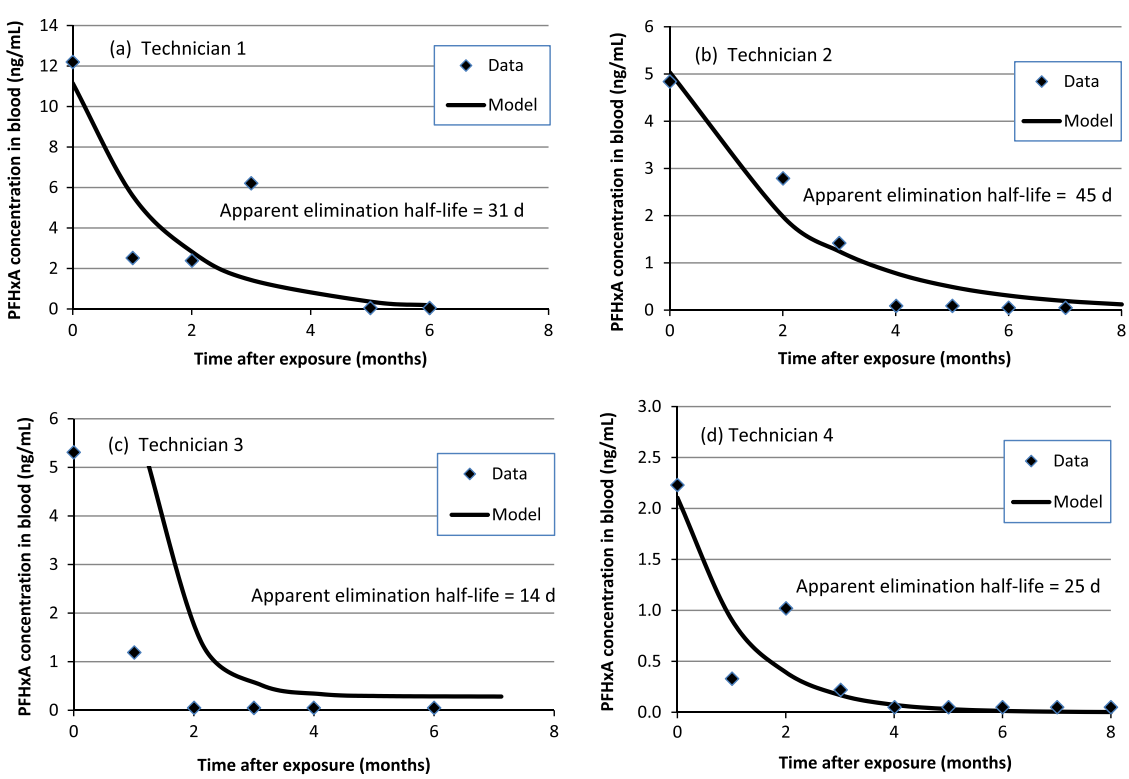

Fig. 1. Representative PFHxA elimination profiles from ski wax technicians.

Table 2
Apparent half-life values of PFHxA in ski wax technicians.

Technician number	Number of samples	Dates of samples	Apparent elimination half-life (d)	Goodness of fit (r^2)
1	6	December 2007–June 2008	31	0.70
2	8	December 2007–August 2008[a]	45	0.94
3	6	February 2008–August 2008	14	1.00
4	9	December 2007–August 2008	25	0.83
6	5	December 2007–June 2008	48	0.98
7	3	December 2007–March 2008	33	0.97
8	5	December 2007–June 2008	49	0.82
	Minimum:		14	
	Geomean:		32	
	Maximum:		49	

[a] January 2008 value of <0.05 ng mL^{-1} not included in regression; if included, half-life = 33 d and r^2 = 0.64.

7.2. Allometric relationships between species

In contrast with long-chain perfluorocarboxylic acids such as PFOA which have distinctly longer elimination half-lives in humans as well as significant half-life differences between mammalian species, the half-life of PFHxA is consistent across mammalian species (human, monkey, rat and mouse) when the results are normalized as a function of body weight.

The conventional approach to evaluate interspecies pharmacokinetic relationships is to use an allometric equation based on a physical parameter such as body weight (Mahmood, 1998) such as $P = A * W^B$ where P is the parameter of interest, W is body weight and A and B are the regression coefficient and exponent of the allometric equation. When human PFHxA blood elimination half-life data was combined with mouse, rat and monkey half-life data and regressed as a function of body weight, the following log-transformed allometric equation was determined:

$$\text{Log(half-life)} = \log(9.54) + 0.93\log(W)$$

where half-life values are in hours and body weight is in kg. The results of this regression are shown for mouse, rat, monkey and human elimination half-lives in Fig. 2.

Typically, chemical elimination half-lives are poorly correlated with body weight due in part to differing volumes of distribution in the various species (Mahmood, 1998). In the regression performed for PFHxA, elimination half-lives were derived from the terminal elimination phase (i.e., after completion of the distribution phase). The resulting regression for PFHxA provides a reasonable fit to the experimental data (r^2 = 0.81) and the elimination half-life value in each species approaches direct proportionality with body weight suggesting a similar volume of distribution in each species. No species-specific factors such as organic ion transporters in the kidney appear to influence the elimination of PFHxA in contrast to the elimination behavior observed for some longer-chain perfluorocarboxylic acids (Weaver et al., 2010; Yang et al., 2010). In addition, no significant gender differences were noted in the elimination half-lives observed in the mouse, rat and monkey.

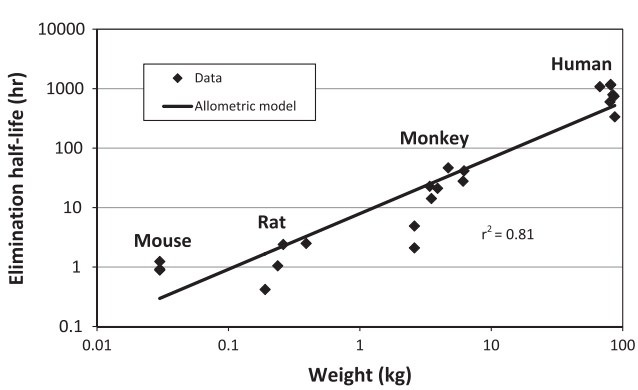

Fig. 2. Allometric correlation of the PFHxA elimination half-life for mouse, rat, monkey and human.

7.3. Toxicological characteristics of PFHxA

Toxicologic evaluations of PFHxA and related salts indicate that this perfluorocarboxylic acid has a favorable toxicology profile (Chengelis et al., 2009b; Loveless et al., 2009). Sodium perfluorohexanoate (NaPFHx) has an acute LC50 of >1750 mg kg^{-1}, is not mutagenic in a bacterial reverse mutation (Ames) assay, does not result in chromosomal aberrations in human lymphocytes treated *in vitro*, and does not present a reproductive, developmental, or neurobehavioral hazard. The no observed adverse effect level (NOAEL) following 90 d of dosing with NaPFHx was 20 mg kg^{-1} d^{-1}, based on nasal lesions in male rats. Benchmark dose (BMD) analysis determined that 13 mg kg^{-1} d^{-1} is the BMDL10 (95% lower confidence limit of a dose resulting in a 10% increase in risk).

Human exposure guidance values normally include a default uncertainty value of 100 composed of a factor of 10 for interspecies differences and another factor of 10 for interindividual differences (IPCS, 2005). The uncertainty factor for interspecies differences is subdivided into a factor of 4.0 for uncertainty in toxicokinetic differences (AK_{UF}) and 2.5 for toxicodynamic uncertainties (AD_{UF}). Based on the direct proportionality of the rate of PFHxA elimination with elimination half-life (i.e., improved kinetic predictability) and the significantly more favorable PFHxA clearance rate, it may be appropriate to reduce the toxicokinetic uncertainty factor (AK_{UF}) from the default value of 4.0 to a lower value which would result in an overall uncertainty factor less than 100.

Using a simple one compartment pharmacokinetic model with conservative assumptions for uptake and volume of distribution, the BMDL10 dose corresponds to blood concentrations of 2000–38 000 ng mL^{-1} based on the observed range of observed elimination rates in rats (see SM for details). For comparison, the blood concentrations of PFHxA observed in human biomonitoring are typically less than 0.1 ng mL^{-1} with few measurements exceeding 1 ng mL^{-1} (Table 1). As a result, even when pharmacokinetic uncertainty factors are not reduced, human exposure to PFHxA is many orders of magnitude lower than concentrations which present a negligible toxicological risk.

8. Conclusions

In large-scale biomonitoring of the general population, PFHxA is detected at relatively low concentrations with median values that are typically less than LOQ values of 0.05–0.10 ng mL^{-1} in serum or plasma. In mammals, PFHxA and its salts are rapidly absorbed,

distributed and eliminated. In the mouse, rat, monkey and human, PFHxA elimination half-life shows a distinct correlation with body weight resulting in predictable clearance values across species. The consistent weight-normalized blood elimination half-life of PFHxA across multiple mammalian species suggests the absence of species-specific transporters in the kidney which are known to extend the elimination half-life of some long-chain PFCAs. The elimination half-life of PFHxA in the mouse and rat is 1–2 h, in monkeys it is 1–2 d and in humans it is 0.5–1.5 months.

PFHxA is rapidly eliminated compared to long-chain perfluoroalkyl acid homologs. The consistent weight-normalized elimination half-lives and improved clearance values for PFHxA in mammalian species indicates that results obtained from animal models are suitable for establishment of PFHxA benchmark dose and reference dose hazard endpoints for use in human risk assessments.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.chemosphere.2013.08.060.

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